

For lyric only prototypes, we recommend TF-IDF cosine. It combines the corrections needed that matter most when making a song recommendation. Cosine normalization stops longer songs from dominating rankings since it simply has more words. Additionally, it helped minimize songs that were dominating rankings, due to filler words ('the', 'of', 'like', etc.), so that distinct words are the driving cause of the similarity score instead.

The precision@2 scores support the choice of TF-IDF since it scored the highest at 0.205, just slightly ahead of raw cosine at 0.196. Both dot product methods were significantly lower scored (TF-IDF dot product and raw dot product at 0.103 and 0.082, respectively). The gap between the cosine methods and the dot product methods was larger than the gap between the TF-IDF and raw dot product, indicating that correcting the song length mattered for the data set more than reweighing individual words. However, TF-IDF cosine still performed better, thus making it the better default recommendation for lyric-only prototype.

When working with cosine similarity (part 3), the raw dot product top match was 'We Gonna Rock'. This is a punk rock song with significantly more total words than the query. Additionally, the query was not a punk genre. While genre is not an end-all factor, it is definitely something to consider in the recommendation. Once corrected for length, cosine similarities top match was 'We Purpose', which from a genre standpoint came much closer, and subjectively sounded closer to the query. Another example comes from the TF-IDF table (part 4). Common filler words like 'you', 'me', and 'that' had some of the lowest IDF weights and appeared in nearly all songs. On the other hand, "strange" or foreign words like 'doo' and 'ven' had IDF weights on the higher end. This shows that reweighting helps center the focus on words that truly distinguish a song from another.

One important limitation to consider is that lyric words count the words that appear but not how it's sung/what it sounds like in the song. Two songs that are drastically different can share very similar vocabulary. Tempo, mood, instruments, and delivery can drastically change a song and can contribute to a song recommended beyond simply what words are included. Listening behavior data would be helpful to fine tune recommendations. This data could be combined with the lyric word count data to then blend similarity scores so that the model recommendations reflect shared sound and audience, not just shared vocabulary.